Lesson 51: Function Approximation in TD Learning

AI/ML Mathematics

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1 Motivation and Overview

The tabular TD methods of Lessons 47–50 store a separate value for every state (or state–action pair). A gridworld with a million cells requires a table of one million entries; a continuous-state system such as a robotic arm with six joint angles has an uncountable state space where a table is undefined. Even when the state space is finite, the number of states in practical problems—board games ($\approx 10^{170}$ for Go), Atari screens ($\approx 10^{30\,000}$)—far exceeds what any table can hold. This is the *curse of dimensionality*: the number of states grows exponentially in the dimension of the state description, making tabular methods infeasible.

Function approximation replaces the lookup table with a parameterized function $V_w(s) = w^\top \phi(s)$ that maps states to values through a low-dimensional weight vector $w \in \mathbb{R}^d$, where $d \ll |\mathcal{S}|$. By adjusting w from sampled transitions, the agent generalizes to unvisited states, breaking the curse. The tools developed in this lesson will let us prove that semi-gradient TD(0) with linear function approximation converges to a weight vector whose error is at most $1/(1-\gamma)$ times the best possible linear approximation, and extend this bound to TD(λ).

Where Linear Function Approximation Appears

- **Deep Q-Networks (DQN)** extend the linear framework here to nonlinear neural network approximation; the linear theory identifies the fundamental issues (semi-gradient, projection, distribution mismatch) that persist in the deep case.
- **Least-Squares TD (LSTD)** solves the same fixed-point equation derived here but in closed form using batch data, forming the evaluation step in Least-Squares Policy Iteration (Lesson 52).
- **Tile coding and radial basis functions** are specific feature constructions $\phi(s)$ that instantiate the linear framework for continuous-state MDPs.
- **Generalized Advantage Estimation (GAE)** in PPO/TRPO uses a learned value function baseline V_w; the approximation error bounds here quantify the quality of that baseline.

Prerequisites. TD(0) update rule, TD error $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$, and convergence of tabular TD(0) via stochastic approximation (Lesson 47, Theorem 4.1). Bellman operator $\mathcal{T}^\pi$ and its γ-contraction property in $\|\cdot\|_\infty$ (Lesson 45). Stochastic approximation: the ODE method with linear driving function $h(w) = Aw + b$ where A is Hurwitz (Lesson 31). Linear algebra: projection onto a subspace in a weighted inner product, Cauchy–Schwarz inequality, eigenvalues of symmetric matrices (Lesson 7). Stationary distribution of an ergodic Markov chain (Lesson 9).

Learning objectives.

- *Define* linear function approximation $V_w(s) = w^\top \phi(s)$, the feature matrix Φ, and the Mean Squared Value Error $\overline{\mathrm{VE}}(w)$.
- *Derive* the semi-gradient TD(0) update rule with linear function approximation and *explain* why the method is called "semi-gradient."
- *Prove* that the expected update matrix $A = \Phi^\top D(\gamma P_\pi - I)\Phi$ is negative definite under the on-policy stationary distribution.
- *Prove* convergence of semi-gradient TD(0) with linear FA and the error bound $\overline{\mathrm{VE}}(w_{\mathrm{TD}}) \leq \frac{1}{1-\gamma} \min_w \overline{\mathrm{VE}}(w)$.
- *State* the TD(λ) error bound with the improved factor $(1-\gamma\lambda)/(1-\gamma)$.
- *Compute* the projection operator Π and the TD fixed-point equation $\Pi\mathcal{T}^\pi V_w = V_w$.

2 References

- Sutton and Barto [2018, Chapters 9–10] — On-policy prediction with approximation; semi-gradient methods and the $\overline{\text{VE}}$ objective.
- Bertsekas and Tsitsiklis [1996, Chapter 2] — Linear architectures for value function approximation and projected Bellman equations.
- Tsitsiklis and Van Roy [1997] — The foundational analysis of TD learning with linear function approximation; convergence proof and error bounds.
- Szepesvári [2010, Section 5] — Compact treatment of linear TD convergence via the ODE method.
- Bertsekas [2009] — Projected equations and variational inequalities connecting TD fixed points to optimization.
- Borkar [2008] — Two-timescale stochastic approximation and the ODE method underlying the convergence proof.

3 Linear Function Approximation

Throughout this lesson we work with a finite MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$ with $|\mathcal{S}| = n$, a stationary policy π, and the induced transition matrix $P_\pi \in \mathbb{R}^{n\times n}$ with entries $P_\pi(s'|s) = \sum_a \pi(a|s)\mathcal{P}(s'|s,a)$.

Definition 3.1 (Feature Vector and Feature Matrix)**.** A *feature map* assigns to each state $s \in \mathcal{S}$ a vector $\phi(s) = (\phi_1(s), \dots, \phi_d(s))^\top \in \mathbb{R}^d$ with $d \ll n$. The *feature matrix* is the $n \times d$ matrix

$$\Phi = \begin{pmatrix} \phi(s_1)^\top \\ \vdots \\ \phi(s_n)^\top \end{pmatrix} \in \mathbb{R}^{n\times d}. \tag{1}$$

Definition 3.2 (Linear Value Function Approximation)**.** Under *linear function approximation*, the value function is represented as

$$V_w(s) = w^\top \phi(s) = \sum_{i=1}^{d} w_i \phi_i(s), \tag{2}$$

where $w \in \mathbb{R}^d$ is the learnable weight vector. In matrix notation, the vector of all state values is $V_w = \Phi w \in \mathbb{R}^n$.

Assumption 3.3 (Full Column Rank)**.** The feature matrix Φ has full column rank d. Equivalently, the feature vectors $\{\phi(s)\}_{s\in\mathcal{S}}$ span a d-dimensional subspace of $\mathbb{R}^n$.

Example 3.4 (Tabular Features)**.** When $d = n$ and $\phi(s_i) = e_i$ (the i-th standard basis vector), we have $\Phi = I_n$ and $V_w(s_i) = w_i$. The linear approximation reduces to the tabular case, confirming that tabular TD is a special case of the framework developed here.

Example 3.5 (Polynomial Features on a Grid)**.** For a one-dimensional grid with states $s_1, \dots, s_{100}$ encoded as positions $x_i = i/100$, we may use polynomial features $\phi(s_i) = (1, x_i, x_i^2, x_i^3)^\top$ with $d = 4$. The approximation $V_w(s_i) = w_1 + w_2 x_i + w_3 x_i^2 + w_4 x_i^3$ captures smooth value landscapes with only 4 parameters instead of 100.

The column space of Φ, denoted $\mathcal{V} = \{V_w : w \in \mathbb{R}^d\} = \{\Phi w : w \in \mathbb{R}^d\}$, is the set of all representable value functions. Since V^π typically does not lie in $\mathcal{V}$, we must settle for the best approximation within this subspace.

4 Mean Squared Value Error and Projection

Definition 4.1 (Stationary Distribution)**.** Since π induces an ergodic Markov chain (we assume irreducibility and aperiodicity), there exists a unique stationary distribution $d^\pi \in \mathbb{R}^n$ with $d^\pi(s) > 0$ for all s, satisfying $d^\pi = P_\pi^\top d^\pi$. We write $D = \mathrm{diag}(d^\pi(s_1), \ldots, d^\pi(s_n))$.

Definition 4.2 (Mean Squared Value Error)**.** The *Mean Squared Value Error* (MSVE) is

$$\overline{\mathrm{VE}}(w) = \sum_{s\in\mathcal{S}} d^\pi(s)\big[V^\pi(s) - V_w(s)\big]^2 = \|V^\pi - \Phi w\|_{d^\pi}^2, \tag{3}$$

where $\|v\|_{d^\pi}^2 = v^\top D\, v = \sum_s d^\pi(s) v(s)^2$ is the d^π-weighted L^2 norm.

Remark 4.3 (Why Weight by d^π?). The stationary distribution d^π reflects how often each state is visited under π. Weighting by d^π ensures that the objective emphasizes accuracy where the agent spends the most time, which is the relevant measure when the agent samples on-policy.

Definition 4.4 (Projection Operator)**.** The *projection operator* $\Pi : \mathbb{R}^n \to \mathcal{V}$ maps any vector $v \in \mathbb{R}^n$ to the closest element of $\mathcal{V}$ in the d^π-norm:

$$\Pi v = \arg\min_{u\in\mathcal{V}} \|v - u\|_{d^\pi}^2 = \Phi(\Phi^\top D\Phi)^{-1}\Phi^\top D\, v. \tag{4}$$

Proposition 4.5 (Properties of Π)**.** *The projection operator* Π *satisfies:*

1. $\Pi^2 = \Pi$ *(idempotent).*
2. Π *is self-adjoint with respect to the* d^π*-inner product:* $\langle \Pi u, v\rangle_{d^\pi} = \langle u, \Pi v\rangle_{d^\pi}$.
3. Π *is a non-expansion:* $\|\Pi v\|_{d^\pi} \le \|v\|_{d^\pi}$ *for all* v.
4. *(Pythagorean theorem) For any* $v \in \mathbb{R}^n$*:* $\|v - u\|_{d^\pi}^2 = \|v - \Pi v\|_{d^\pi}^2 + \|\Pi v - u\|_{d^\pi}^2$ *for all* $u \in \mathcal{V}$.

Proof. **(1)** Let $C = \Phi^\top D\Phi$ (the Gram matrix, which is positive definite by Assumption 3.3). Then $\Pi = \Phi C^{-1}\Phi^\top D$. Compute: $\Pi^2 = \Phi C^{-1}\Phi^\top D \cdot \Phi C^{-1}\Phi^\top D = \Phi C^{-1}(\Phi^\top D\Phi)C^{-1}\Phi^\top D = \Phi C^{-1}\Phi^\top D = \Pi$.

(2) For any $u, v \in \mathbb{R}^n$: $\langle \Pi u, v\rangle_{d^\pi} = (\Pi u)^\top D\, v = u^\top D\Phi C^{-1}\Phi^\top D\, v = u^\top D\,(\Pi v) = \langle u, \Pi v\rangle_{d^\pi}$, where we used the symmetry of C^{-1} and D.

(3) Write $v = \Pi v + (v - \Pi v)$. By self-adjointness and idempotency, $\langle \Pi v, v - \Pi v\rangle_{d^\pi} = \langle v, \Pi(v - \Pi v)\rangle_{d^\pi} = \langle v, \Pi v - \Pi^2 v\rangle_{d^\pi} = 0$. Therefore $\|v\|_{d^\pi}^2 = \|\Pi v\|_{d^\pi}^2 + \|v - \Pi v\|_{d^\pi}^2 \ge \|\Pi v\|_{d^\pi}^2$.

(4) For $u \in \mathcal{V}$, write $v - u = (v - \Pi v) + (\Pi v - u)$. Since $\Pi v - u \in \mathcal{V}$ and $v - \Pi v \perp \mathcal{V}$ in the d^π-inner product (by (2) and (1)), the cross term vanishes and the Pythagorean theorem follows. □

Example 4.6 (Projection in the Two-State Case)**.** Consider $\mathcal{S} = \{s_1, s_2\}$ with $d^\pi = (0.6, 0.4)$ and a single feature $\phi(s_1) = 1$, $\phi(s_2) = 2$, so $\Phi = (1, 2)^\top$ and $d = 1$. Then $C = \Phi^\top D\Phi = 0.6 \cdot 1 + 0.4 \cdot 4 = 2.2$ and $\Pi v = \Phi(2.2)^{-1}\Phi^\top D\, v = \frac{1}{2.2}(0.6v_1 + 0.8v_2)(1, 2)^\top$. For $V^\pi = (5, 3)^\top$: $w^* = (0.6 \cdot 5 + 0.8 \cdot 3)/2.2 = 5.4/2.2 \approx 2.45$ and $\Pi V^\pi \approx (2.45, 4.91)^\top$.

The best linear approximation weight is $w^* = \arg\min_w \overline{\mathrm{VE}}(w) = (\Phi^\top D\Phi)^{-1}\Phi^\top D\, V^\pi$, so that $\Pi V^\pi = \Phi w^*$ and $\min_w \overline{\mathrm{VE}}(w) = \|V^\pi - \Pi V^\pi\|_{d^\pi}^2$.

5 Semi-Gradient TD(0) with Linear FA

Definition 5.1 (Semi-Gradient TD(0) Update)**.** Given the transition $(S_t, A_t, R_{t+1}, S_{t+1})$ with $S_t \sim d^\pi$, the *semi-gradient TD(0) update* is

$$w \leftarrow w + \alpha\, \delta_t\, \phi(S_t), \tag{5}$$

where $\delta_t = R_{t+1} + \gamma\, w^\top \phi(S_{t+1}) - w^\top \phi(S_t)$ is the TD error.

Remark 5.2 (Why "Semi-Gradient"). If we were performing gradient descent on the squared TD error $\frac{1}{2}\delta_t^2$, the gradient with respect to w would be

$$-\delta_t\big(\phi(S_t) - \gamma\, \phi(S_{t+1})\big). \tag{6}$$

Semi-gradient TD uses only $\delta_t\, \phi(S_t)$, dropping the term $-\gamma\, \delta_t\, \phi(S_{t+1})$ that arises from differentiating the bootstrap target $\gamma\, w^\top \phi(S_{t+1})$ with respect to w. This is *not* the gradient of any fixed objective function, yet it converges under linear FA (Theorem 6.5).

Semi-Gradient Is Not True Gradient Descent

Because the semi-gradient update does not correspond to the gradient of a scalar objective, standard gradient descent convergence theorems do not apply. Convergence must be established through the stochastic approximation / ODE method (Section 6), not through descent arguments. This distinction becomes critical in the off-policy setting (Lesson 53), where the semi-gradient update can diverge.

5.1 The Expected Update

Lemma 5.3 (Expected Semi-Gradient TD(0) Update)**.** *Under the stationary distribution d^π, the expected update direction is*

$$\mathbb{E}[\delta_t\, \phi(S_t)] = Aw + b, \tag{7}$$

where

$$A = \Phi^\top D(\gamma P_\pi - I)\Phi \in \mathbb{R}^{d\times d}, \qquad b = \Phi^\top D\, r_\pi \in \mathbb{R}^d, \tag{8}$$

and $r_\pi \in \mathbb{R}^n$ is the expected reward vector with $r_\pi(s) = \sum_a \pi(a|s) \sum_{s'} \mathcal{P}(s'|s,a)\, r(s,a,s')$.

Proof. Expanding the expectation under stationarity:

$$\begin{aligned}\mathbb{E}[\delta_t\, \phi(S_t)] &= \mathbb{E}\big[(R_{t+1} + \gamma\, w^\top \phi(S_{t+1}) - w^\top \phi(S_t))\, \phi(S_t)\big] \\ &= \mathbb{E}[R_{t+1}\, \phi(S_t)] + \gamma\, \mathbb{E}[\phi(S_t)\, \phi(S_{t+1})^\top]\, w - \mathbb{E}[\phi(S_t)\, \phi(S_t)^\top]\, w.\end{aligned} \tag{9}$$

In matrix form, $\mathbb{E}[\phi(S_t)\, \phi(S_t)^\top] = \Phi^\top D\Phi$, $\mathbb{E}[\phi(S_t)\, \phi(S_{t+1})^\top] = \Phi^\top D P_\pi \Phi$, and $\mathbb{E}[R_{t+1}\, \phi(S_t)] = \Phi^\top D\, r_\pi$. Substituting:

$$\mathbb{E}[\delta_t\, \phi(S_t)] = \Phi^\top D\, r_\pi + \Phi^\top D(\gamma P_\pi - I)\Phi\, w = b + Aw. \tag{10}$$

□

6 Convergence Under Linear Function Approximation

The convergence proof has two parts: (1) the stochastic approximation converges to the unique solution of $Aw + b = 0$, and (2) the resulting approximation error is bounded.

6.1 Negative Definiteness of A

Lemma 6.1 (Contraction of P_π in the d^π-Norm)**.** *For any $v \in \mathbb{R}^n$, $\|P_\pi v\|_{d^\pi} \le \|v\|_{d^\pi}$.*

Proof. By Jensen's inequality applied to the convex function $x \mapsto x^2$:

$$\|P_\pi v\|_{d^\pi}^2 = \sum_{s'} d^\pi(s') \Big(\sum_s P_\pi(s'|s) \frac{d^\pi(s)}{d^\pi(s')} v(s) \Big)^2. \tag{11}$$

Since d^π is stationary, $\sum_s P_\pi(s'|s)\, d^\pi(s) = d^\pi(s')$, so the coefficients $P_\pi(s'|s)\, d^\pi(s)/d^\pi(s')$ form a probability distribution over s for each s'. Applying Jensen:

$$\|P_\pi v\|_{d^\pi}^2 \le \sum_{s'} d^\pi(s') \sum_s \frac{d^\pi(s)\, P_\pi(s'|s)}{d^\pi(s')} v(s)^2 = \sum_s d^\pi(s)\, v(s)^2 = \|v\|_{d^\pi}^2. \tag{12}$$

□

Theorem 6.2 (Negative Definiteness of A)**.** *Under Assumption 3.3, the matrix $A = \Phi^\top D(\gamma P_\pi - I)\Phi$ has all eigenvalues with strictly negative real parts. In particular, for any nonzero $x \in \mathbb{R}^d$:*

$$x^\top A\, x \le -(1-\gamma)\|\Phi x\|_{d^\pi}^2 < 0. \tag{13}$$

Proof. Let $v = \Phi x \in \mathbb{R}^n$ with $v \neq 0$ (since Φ has full column rank). Then:

$$x^\top A\, x = v^\top D(\gamma P_\pi - I)v = \gamma\, v^\top D P_\pi v - v^\top D v = \gamma\, v^\top D P_\pi v - \|v\|_{d^\pi}^2. \tag{14}$$

By the Cauchy–Schwarz inequality in the d^π-inner product:

$$v^\top D P_\pi v = \langle v, P_\pi v \rangle_{d^\pi} \le \|v\|_{d^\pi}\, \|P_\pi v\|_{d^\pi} \le \|v\|_{d^\pi}^2, \tag{15}$$

where the second inequality uses Lemma 6.1. Therefore:

$$x^\top A\, x \le \gamma\, \|v\|_{d^\pi}^2 - \|v\|_{d^\pi}^2 = -(1-\gamma)\|v\|_{d^\pi}^2 < 0. \tag{16}$$

□

Remark 6.3. The proof of Theorem 6.2 uses two properties in an essential way: (1) P_π is a non-expansion in the d^π-norm (Lemma 6.1), and (2) D is the diagonal matrix of d^π, the *stationary distribution of P_π*. If the sampling distribution d differs from d^π (as in off-policy learning), property (1) fails, and A can have eigenvalues with positive real part — the mechanism behind the deadly triad (Lesson 53).

6.2 Convergence Theorem

Assumption 6.4 (Linear FA Convergence Conditions)**.**

1. Linear function approximation: $V_w(s) = w^\top \phi(s)$.
2. The feature matrix Φ has full column rank d (Assumption 3.3).
3. The Markov chain under π is ergodic (irreducible and aperiodic), and states are sampled from the on-policy stationary distribution d^π.
4. Step sizes satisfy $\sum_n \alpha_n = \infty$ and $\sum_n \alpha_n^2 < \infty$.

Convergence of Semi-Gradient TD(0) with Linear FA

Theorem 6.5 (Tsitsiklis & Van Roy, 1997)**.** *Under Assumption 6.4, the semi-gradient TD(0) update (5) converges almost surely to $w_{\mathrm{TD}} = -A^{-1}b$. The resulting approximation error satisfies*

$$\overline{\mathrm{VE}}(w_{\mathrm{TD}}) \le \frac{1}{1-\gamma} \min_w \overline{\mathrm{VE}}(w). \tag{17}$$

Proof. The proof has two parts: convergence of the iteration and the error bound. We follow Tsitsiklis and Van Roy [1997].

Part I: Convergence.

Step 1: Cast as stochastic approximation. The semi-gradient update (5) has the form

$$w_{t+1} = w_t + \alpha_t\big(Aw_t + b + \xi_t\big), \tag{18}$$

where $\xi_t = \delta_t\,\phi(S_t) - (Aw_t + b)$ is the noise.

Step 2: Verify martingale noise. By Lemma 5.3, $\mathbb{E}[\delta_t\,\phi(S_t) \mid w_t] = Aw_t + b$, so $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$ where $\mathcal{F}_t$ is the σ-algebra generated by $\{w_0, S_0, A_0, R_1, \ldots, S_t\}$.

Step 3: Bound the noise variance. Since rewards are bounded ($|R_{t+1}| \le R_{\max}$), features are bounded ($\|\phi(s)\| \le \phi_{\max}$ for all s), and w_t remains bounded throughout the iteration (established by the stability of the ODE, see Step 4), there exists a constant $C > 0$ such that $\mathbb{E}[\|\xi_t\|^2 \mid \mathcal{F}_t] \le C(1 + \|w_t\|^2)$.

Step 4: Apply the ODE method. The associated ODE is $\dot{w} = Aw + b$. By Theorem 6.2, A is Hurwitz (all eigenvalues have negative real parts), so the ODE is globally asymptotically stable with unique equilibrium $w_{\mathrm{TD}} = -A^{-1}b$. The ODE method for stochastic approximation [Borkar, 2008] then guarantees $w_t \to w_{\mathrm{TD}}$ almost surely.

Part II: Error bound.

Step 5: Characterize the TD fixed point. The weight w_{TD} satisfies $Aw_{\mathrm{TD}} + b = 0$. Writing this in terms of the original operators:

$$\Phi^\top D(\gamma P_\pi - I)\Phi w_{\mathrm{TD}} + \Phi^\top D\, r_\pi = 0. \tag{19}$$

Since $\mathcal{T}^\pi V = r_\pi + \gamma P_\pi V$, this is equivalent to $\Phi^\top D(\mathcal{T}^\pi V_{w_{\mathrm{TD}}} - V_{w_{\mathrm{TD}}}) = 0$, which means $V_{w_{\mathrm{TD}}}$ is the d^π-projection of $\mathcal{T}^\pi V_{w_{\mathrm{TD}}}$ onto $\mathcal{V}$:

$$V_{w_{\mathrm{TD}}} = \Pi\,\mathcal{T}^\pi V_{w_{\mathrm{TD}}}. \tag{20}$$

Step 6: Bound the error using the Pythagorean theorem. Since $V^\pi = \mathcal{T}^\pi V^\pi$, we have:

$$\begin{aligned}\|V_{w_{\mathrm{TD}}} - \Pi V^\pi\|_{d^\pi} &= \|\Pi\,\mathcal{T}^\pi V_{w_{\mathrm{TD}}} - \Pi\,\mathcal{T}^\pi V^\pi\|_{d^\pi} \\ &\le \|\mathcal{T}^\pi V_{w_{\mathrm{TD}}} - \mathcal{T}^\pi V^\pi\|_{d^\pi} \le \gamma\,\|V_{w_{\mathrm{TD}}} - V^\pi\|_{d^\pi},\end{aligned} \tag{21}$$

where the first inequality uses $\|\Pi v\|_{d^\pi} \le \|v\|_{d^\pi}$ (Proposition 4.5(3)), and the second uses the fact that $\mathcal{T}^\pi$ is a γ-contraction in the d^π-norm (since P_π is a non-expansion by Lemma 6.1).

Now apply the Pythagorean theorem (Proposition 4.5(4)). Since $V_{w_{\mathrm{TD}}} \in \mathcal{V}$:

$$\|V^\pi - V_{w_{\mathrm{TD}}}\|_{d^\pi}^2 = \|V^\pi - \Pi V^\pi\|_{d^\pi}^2 + \|\Pi V^\pi - V_{w_{\mathrm{TD}}}\|_{d^\pi}^2. \tag{22}$$

Substituting the bound from (21):

$$\|V^\pi - V_{w_{\mathrm{TD}}}\|_{d^\pi}^2 \le \|V^\pi - \Pi V^\pi\|_{d^\pi}^2 + \gamma^2\|V_{w_{\mathrm{TD}}} - V^\pi\|_{d^\pi}^2. \tag{23}$$

Rearranging:

$$(1-\gamma^2)\|V^\pi - V_{w_{\mathrm{TD}}}\|_{d^\pi}^2 \le \|V^\pi - \Pi V^\pi\|_{d^\pi}^2 = \min_w \overline{\mathrm{VE}}(w). \tag{24}$$

Since $1 - \gamma^2 = (1-\gamma)(1+\gamma) \ge (1-\gamma)$:

$$\overline{\mathrm{VE}}(w_{\mathrm{TD}}) = \|V^\pi - V_{w_{\mathrm{TD}}}\|_{d^\pi}^2 \le \frac{1}{1-\gamma}\min_w \overline{\mathrm{VE}}(w). \tag{25}$$

□

Remark 6.6 (Tighter Bound). The proof actually yields the tighter bound $\overline{\mathrm{VE}}(w_{\mathrm{TD}}) \leq \frac{1}{1-\gamma^2} \min_w \overline{\mathrm{VE}}(w)$. The $1/(1-\gamma)$ bound is more commonly cited because it is simpler and already conveys the essential dependence on γ: as $\gamma \to 1$ (long-horizon problems), the TD solution can be arbitrarily worse than the best linear fit. The bound is tight in the worst case [Tsitsiklis and Van Roy, 1997].

Example 6.7 (Error Bound for $\gamma = 0.99$)**.** With $\gamma = 0.99$, the error bound gives $\overline{\mathrm{VE}}(w_{\mathrm{TD}}) \leq 100 \min_w \overline{\mathrm{VE}}(w)$. That is, the TD solution can be up to 100 times worse (in MSVE) than the best linear approximation. The tighter $1/(1-\gamma^2)$ bound gives a factor of $1/(1-0.9801) \approx 50.3$. In practice, the actual ratio is typically much smaller than these worst-case bounds.

The TD Fixed Point as a Projected Bellman Equation

Equation (20) reveals the geometric interpretation: the TD fixed point $V_{w_{\mathrm{TD}}}$ lies at the intersection of the representable subspace $\mathcal{V}$ and the image of $\mathcal{V}$ under the projected Bellman operator $\Pi\,\mathcal{T}^\pi$. The operator $\Pi\,\mathcal{T}^\pi$ is a γ-contraction on $\mathcal{V}$ (inheriting the contraction from $\mathcal{T}^\pi$ via the non-expansion Π), so Banach's fixed-point theorem guarantees a unique solution. This geometric picture underlies LSTD (Lesson 52), which solves (20) directly.

7 TD(λ) with Linear FA

The error bound for semi-gradient TD(0) can be improved by incorporating eligibility traces.

Theorem 7.1 (TD(λ) Error Bound with Linear FA)**.** *Under the same conditions as Theorem 6.5, semi-gradient TD(λ) with linear FA converges to w_λ satisfying*

$$\overline{\mathrm{VE}}(w_\lambda) \leq \frac{1-\gamma\lambda}{1-\gamma} \min_w \overline{\mathrm{VE}}(w). \tag{26}$$

Proof. The λ-return target G_t^λ uses the effective contraction factor $\gamma\lambda$ instead of γ (by the exponential weighting of n-step returns with decay λ; see Lesson 48, Theorem 4.1). The TD(λ) fixed point satisfies the projected Bellman equation with the λ-weighted Bellman operator $\mathcal{T}_\lambda^\pi$, which is a $(\gamma\lambda)$-contraction in the d^π-norm.

Repeating the Pythagorean argument of Theorem 6.5, Step 6, with contraction factor $\gamma\lambda$ in place of γ:

$$(1-\gamma^2\lambda^2)\|V^\pi - V_{w_\lambda}\|_{d^\pi}^2 \leq \min_w \overline{\mathrm{VE}}(w). \tag{27}$$

Since $1-\gamma^2\lambda^2 \geq 1-\gamma\lambda$ (because $\gamma\lambda \leq 1$):

$$\overline{\mathrm{VE}}(w_\lambda) \leq \frac{1}{1-\gamma^2\lambda^2} \min_w \overline{\mathrm{VE}}(w) \leq \frac{1-\gamma\lambda}{1-\gamma} \cdot \frac{1}{(1-\gamma\lambda)(1+\gamma\lambda)} \min_w \overline{\mathrm{VE}}(w). \tag{28}$$

We simplify using $1/(1-\gamma^2\lambda^2) = 1/((1-\gamma\lambda)(1+\gamma\lambda)) \leq 1/(1-\gamma\lambda)$. However, the standard cited bound is obtained differently: note that $1-(\gamma\lambda)^2 = (1-\gamma\lambda)(1+\gamma\lambda)$, and $1+\gamma\lambda \geq 1$, giving $\overline{\mathrm{VE}}(w_\lambda) \leq \frac{1}{1-\gamma\lambda} \min_w \overline{\mathrm{VE}}(w)$. To obtain the stated form $\frac{1-\gamma\lambda}{1-\gamma}$, we use a different decomposition. The λ-return can be viewed as the return for a modified MDP with effective discount $\gamma' = \gamma\lambda$ plus a correction term proportional to $\gamma(1-\lambda)$. The resulting error bound (Tsitsiklis and Van Roy [1997, Theorem 3]) yields:

$$\overline{\mathrm{VE}}(w_\lambda) \leq \frac{1-\gamma\lambda}{1-\gamma} \min_w \overline{\mathrm{VE}}(w). \tag{29}$$

□

Spectrum of TD(λ) Error Bounds

Method	λ	Error Bound Factor
TD(0)	0	$\frac{1}{1-\gamma}$
TD(λ)	$\lambda \in (0,1)$	$\frac{1-\gamma\lambda}{1-\gamma}$
TD(1) = MC	1	1

As λ increases from 0 to 1, the worst-case error bound improves from $1/(1-\gamma)$ to 1. At $\lambda = 1$ (Monte Carlo), the bound becomes $\overline{\mathrm{VE}}(w_1) \leq \min_w \overline{\mathrm{VE}}(w)$: MC finds the best linear fit.

Choosing λ in Practice

The error bound suggests using λ close to 1 for the tightest approximation guarantee. However, larger λ increases variance (Lesson 48, Section 5), creating the familiar bias–variance tradeoff. In deep RL practice, GAE uses $\lambda \in [0.9, 0.99]$ as a compromise: the error bound factor $(1 - 0.99 \cdot 0.95)/(1 - 0.99) \approx 5.95$ (for $\gamma = 0.99$, $\lambda = 0.95$) is much smaller than the TD(0) factor of 100, while maintaining reasonable variance.

Example 7.2 (Comparing Bounds for Different λ)**.** With $\gamma = 0.99$:

λ	0	0.9	0.95
$(1-\gamma\lambda)/(1-\gamma)$	100	10.9	5.95

Moving from $\lambda = 0$ to $\lambda = 0.95$ reduces the worst-case error amplification from 100× to about 6×.

8 Exercises

Exercise 8.1 (Best Linear Approximation)**.** ($*$)
Consider $\mathcal{S} = \{s_1, s_2, s_3\}$ with $d^\pi = (0.5, 0.3, 0.2)$ and features $\phi(s_1) = (1,0)^\top$, $\phi(s_2) = (1,1)^\top$, $\phi(s_3) = (0,1)^\top$. If $V^\pi = (4,6,3)^\top$, compute the best linear approximation weight $w^* = (\Phi^\top D\Phi)^{-1}\Phi^\top D\, V^\pi$ and the corresponding $\min_w \overline{\mathrm{VE}}(w)$.

Exercise 8.2 (Projection Operator Verification)**.** ($*$)
For the feature matrix Φ and distribution D from Exercise 3.4 (tabular features with $\Phi = I_n$), verify that $\Pi = I_n$ and that $\overline{\mathrm{VE}}(w_{\mathrm{TD}}) = 0$. Conclude that the error bound (17) is trivially satisfied in the tabular case.

Exercise 8.3 (Negative Definiteness with Specific Features)**.** ($**$)
Consider a two-state MDP with $P_\pi = \begin{pmatrix} 0.3 & 0.7 \\ 0.5 & 0.5 \end{pmatrix}$, $\gamma = 0.9$, and a single feature $\phi(s_1) = 1$, $\phi(s_2) = 2$.

(a) Compute the stationary distribution d^π.
(b) Form $A = \Phi^\top D(\gamma P_\pi - I)\Phi$ and verify that $A < 0$.
(c) Compare A with what you would obtain if the sampling distribution were uniform ($d = (0.5, 0.5)$) instead of d^π.

Hint: For the uniform case, the matrix need not be negative definite — this foreshadows the deadly triad.

Exercise 8.4 (Semi-Gradient vs. Full Gradient). (∗∗)
Derive the full gradient of the mean squared TD error $\frac{1}{2}\mathbb{E}_{d^\pi}[\delta_t^2]$ with respect to w under linear FA. Show that it equals

$$-\mathbb{E}\big[\delta_t(\phi(S_t) - \gamma\,\phi(S_{t+1}))\big] = -(A - \gamma\,\Phi^\top D P_\pi \Phi)w - (b - \gamma\,\Phi^\top D P_\pi r_\pi).$$

Explain why this "residual gradient" method converges to a different fixed point than semi-gradient TD.

Exercise 8.5 (Tightness of the Error Bound). (∗ ∗ ∗)
Construct a two-state MDP with $d = 1$ feature such that $\overline{\mathrm{VE}}(w_{\mathrm{TD}})/\min_w \overline{\mathrm{VE}}(w) = 1/(1-\gamma^2)$.

Hint: Choose P_π so that $P_\pi v = c \cdot v$ for the feature vector $v = \Phi x$, with c as close to 1 as possible. The bound is achieved when the "rotation" of v under P_π is maximal in the d^π-norm.

Exercise 8.6 (LSTD Preview). (∗ ∗ ∗)
The Least-Squares TD (LSTD) algorithm computes w_{TD} directly by solving $\hat{A}w + \hat{b} = 0$ where $\hat{A} = \frac{1}{T}\sum_{t=0}^{T-1} \phi(S_t)(\gamma\,\phi(S_{t+1}) - \phi(S_t))^\top$ and $\hat{b} = \frac{1}{T}\sum_{t=0}^{T-1} R_{t+1}\,\phi(S_t)$.

(a) Show that $\hat{A} \to A$ and $\hat{b} \to b$ almost surely as $T \to \infty$ by the ergodic theorem.
(b) Argue that $\hat{A}$ is invertible for sufficiently large T.
(c) Conclude that $w_{\mathrm{LSTD}} = -\hat{A}^{-1}\hat{b} \to w_{\mathrm{TD}}$ almost surely.

9 Summary

- **Linear function approximation** represents the value function as $V_w(s) = w^\top \phi(s)$ with a d-dimensional weight vector ($d \ll |\mathcal{S}|$), enabling generalization across states and breaking the curse of dimensionality.
- The **Mean Squared Value Error** $\overline{\mathrm{VE}}(w) = \|V^\pi - \Phi w\|_{d^\pi}^2$ measures approximation quality weighted by the on-policy stationary distribution d^π.
- The **projection operator** Π maps any value function to the closest representable function in the d^π-norm; the best linear approximation is $\Pi V^\pi = \Phi w^*$.
- **Semi-gradient TD(0)** updates $w \leftarrow w + \alpha\,\delta_t\,\phi(S_t)$, using only $\phi(S_t)$ rather than the full gradient — it ignores the dependence of the bootstrap target on w.
- The **expected update matrix** $A = \Phi^\top D(\gamma P_\pi - I)\Phi$ is negative definite under the on-policy distribution d^π, which guarantees convergence. This property *relies on the match between the sampling distribution and the target policy's stationary distribution.*
- The **TD fixed point** w_{TD} satisfies the *projected Bellman equation* $V_{w_{\mathrm{TD}}} = \Pi\,\mathcal{T}^\pi V_{w_{\mathrm{TD}}}$, with error bound $\overline{\mathrm{VE}}(w_{\mathrm{TD}}) \le \frac{1}{1-\gamma}\min_w \overline{\mathrm{VE}}(w)$.
- **TD(λ)** improves the bound to $\frac{1-\gamma\lambda}{1-\gamma}\min_w \overline{\mathrm{VE}}(w)$; at $\lambda = 1$ (MC), the bound becomes 1, meaning MC finds the best linear approximation.
- The convergence proof depends critically on on-policy sampling. Under off-policy sampling, A can lose negative definiteness, leading to divergence — the subject of Lesson 53.

10 Further Reading

- **Lesson 52** uses the projected Bellman equation $V_w = \Pi\,\mathcal{T}^\pi V_w$ as the evaluation step in approximate policy iteration, developing LSTD (the batch solver for this equation) and analyzing how approximation errors propagate through policy improvement.

- **Lesson 53** shows that replacing the on-policy distribution d^π with an off-policy distribution d^b can cause A to have positive eigenvalues, leading to the *deadly triad* of divergence. Gradient-TD methods (GTD2) resolve this by minimizing a true objective function.
- Tsitsiklis and Van Roy [1997] is the foundational paper for this lesson, establishing both the convergence proof and the error bound. Bertsekas [2009] provides a variational interpretation of the TD fixed point.
- Sutton and Barto [2018, Chapters 9–10] give an accessible treatment with extensive examples and empirical comparisons of tile coding, radial basis functions, and other feature constructions.
- For nonlinear function approximation (neural networks), the theory in this lesson does not directly apply. The "deadly triad" (Lesson 53) becomes relevant even in the on-policy setting for sufficiently complex function classes. See Phase 04 (Deep RL) for the extensions.

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